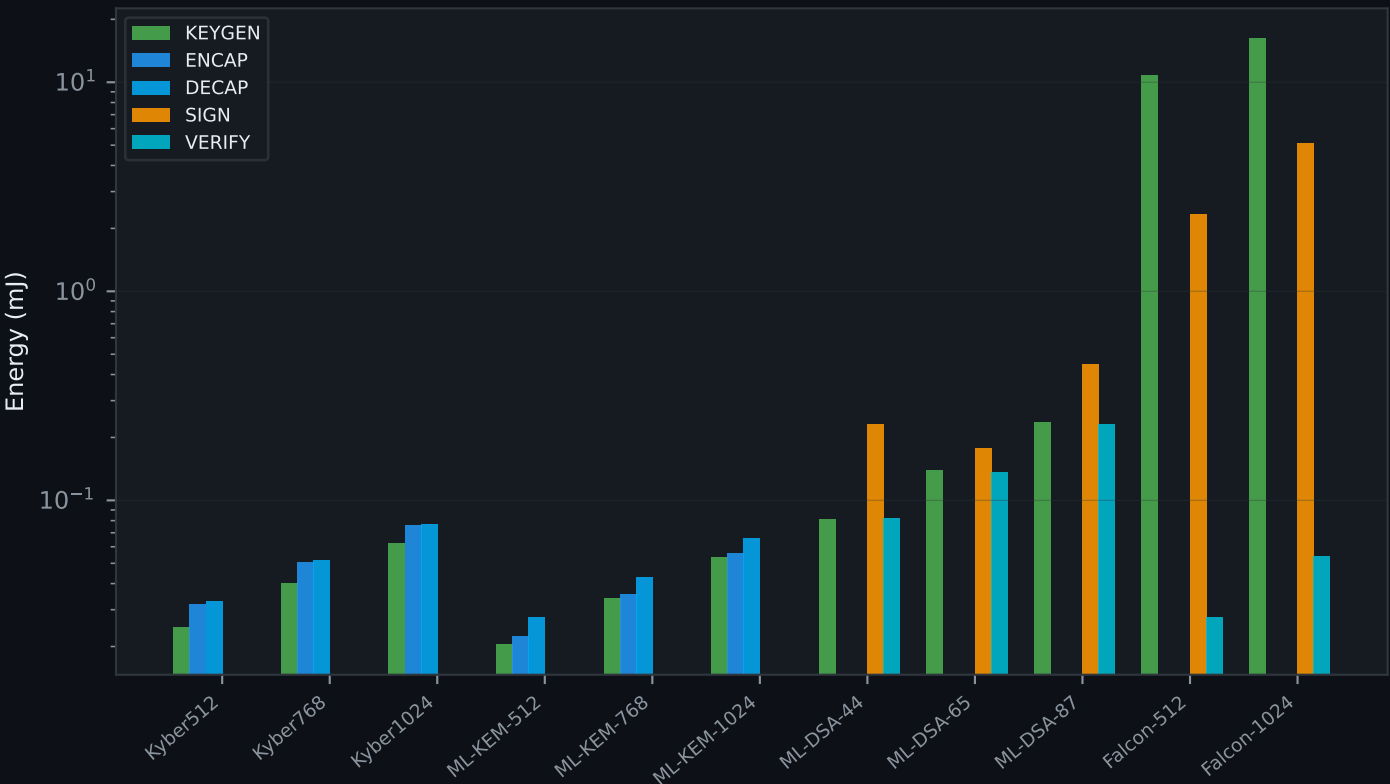
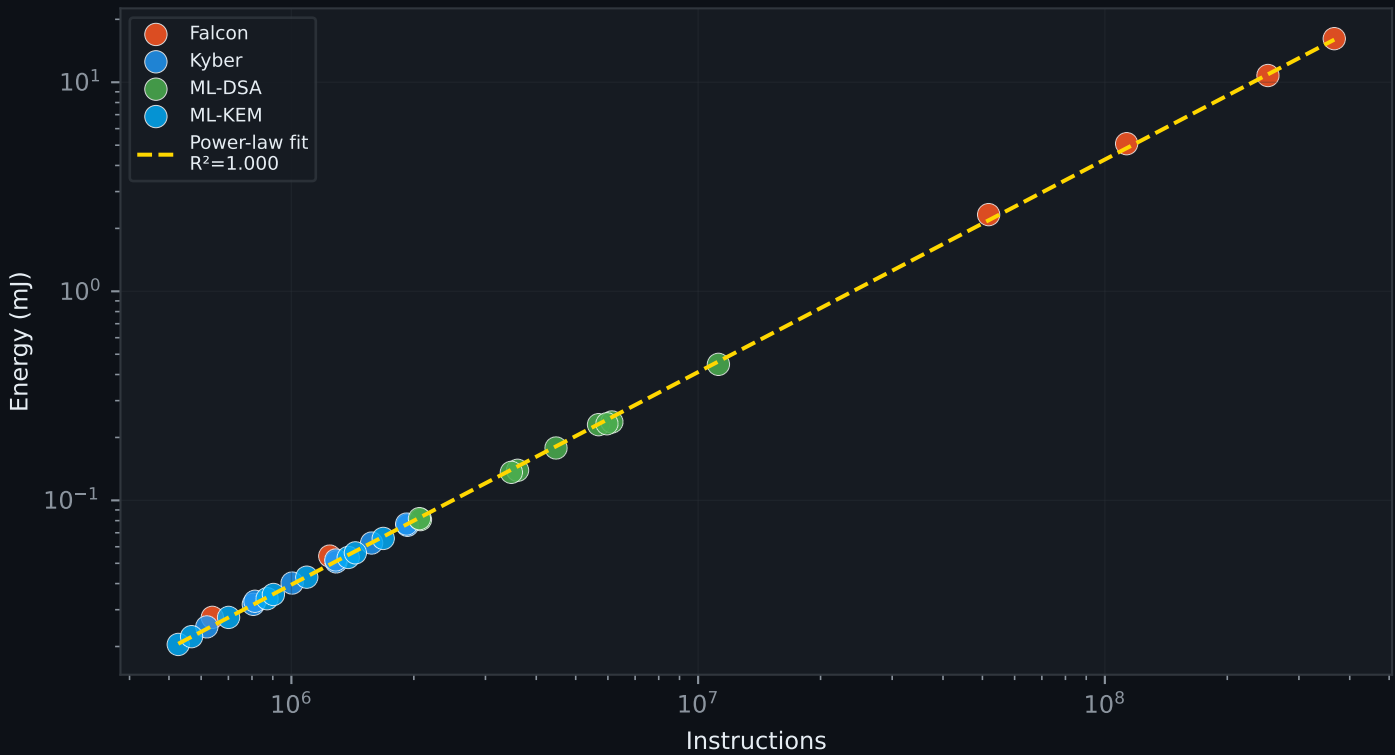


PQC Energy Usage — Real gem5 ARM Measurements
ARM Cortex-M4 | 64 MHz | 66 mW active | liboqs

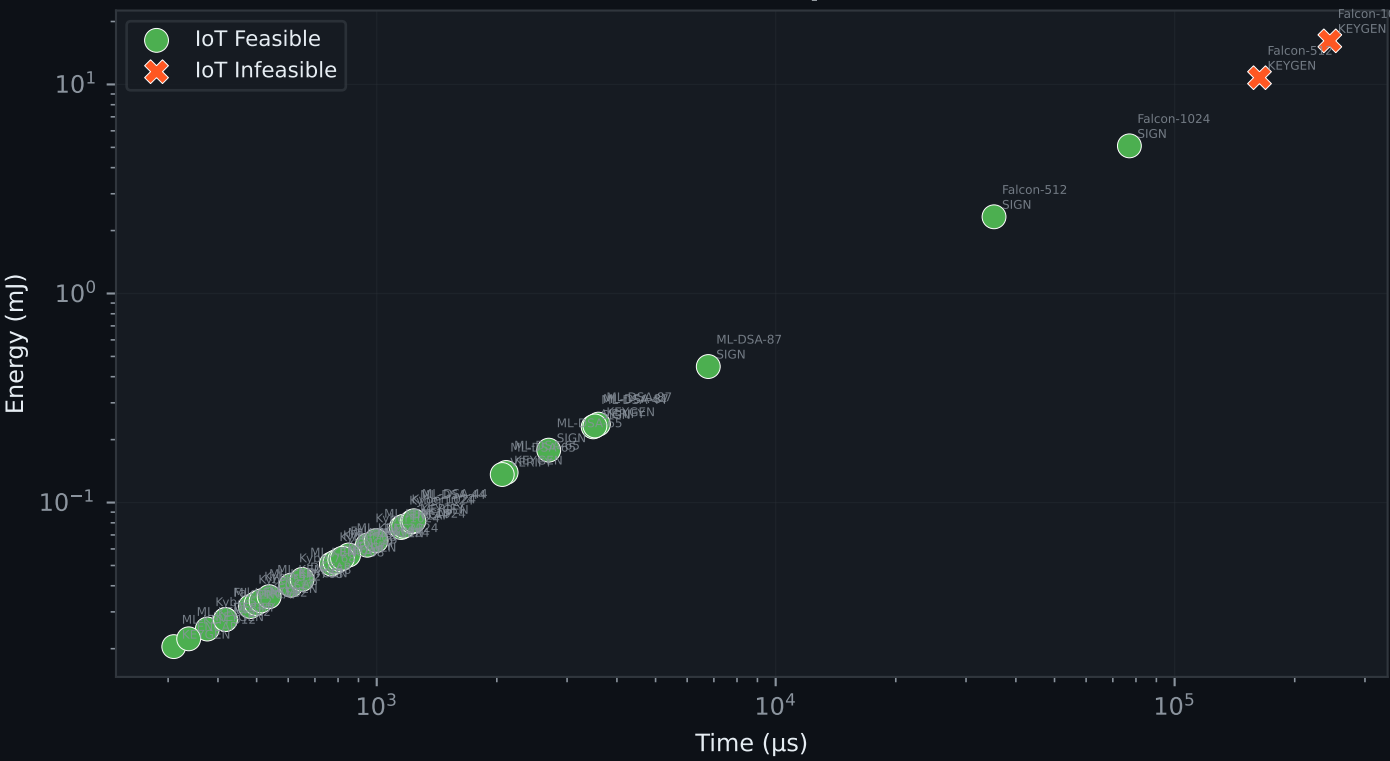
A — Energy per Algorithm & Operation



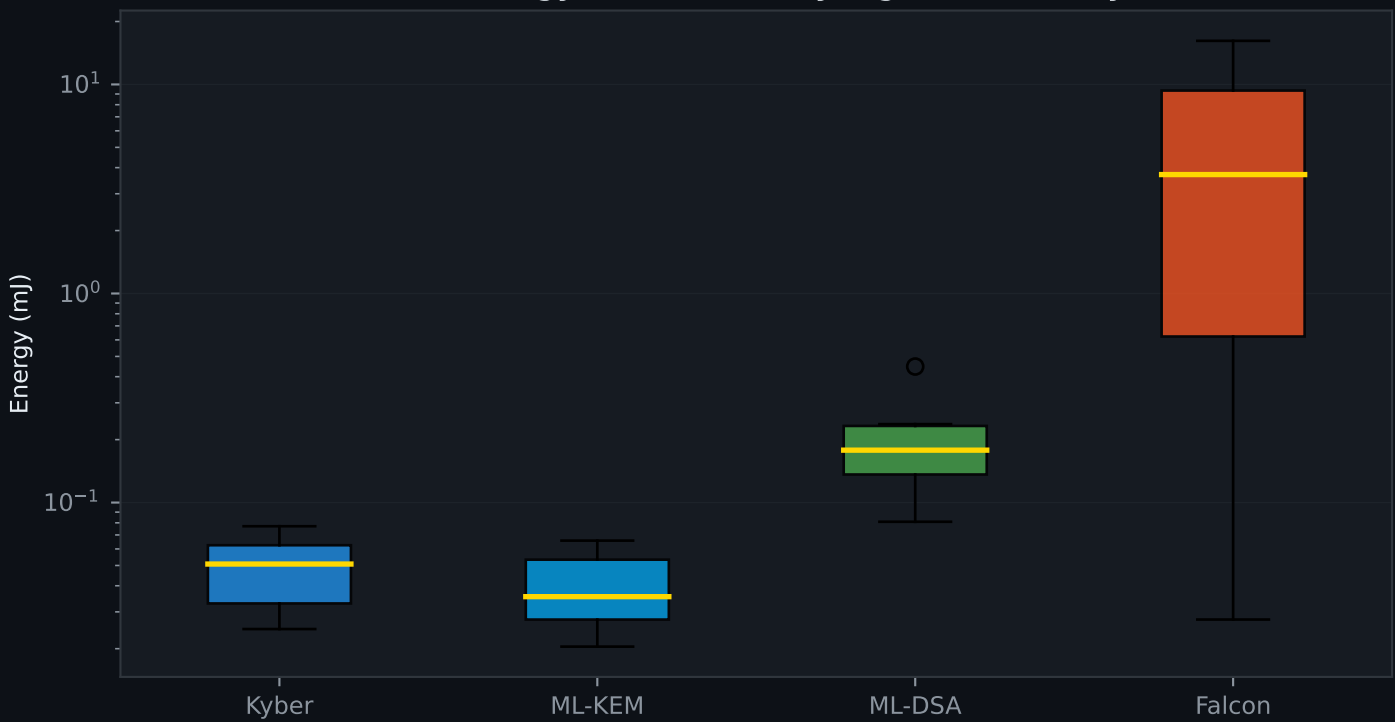
B — Instructions vs Energy (log-log)



C — IoT Feasibility Map
(KEYGEN <10ms, ops <100ms)

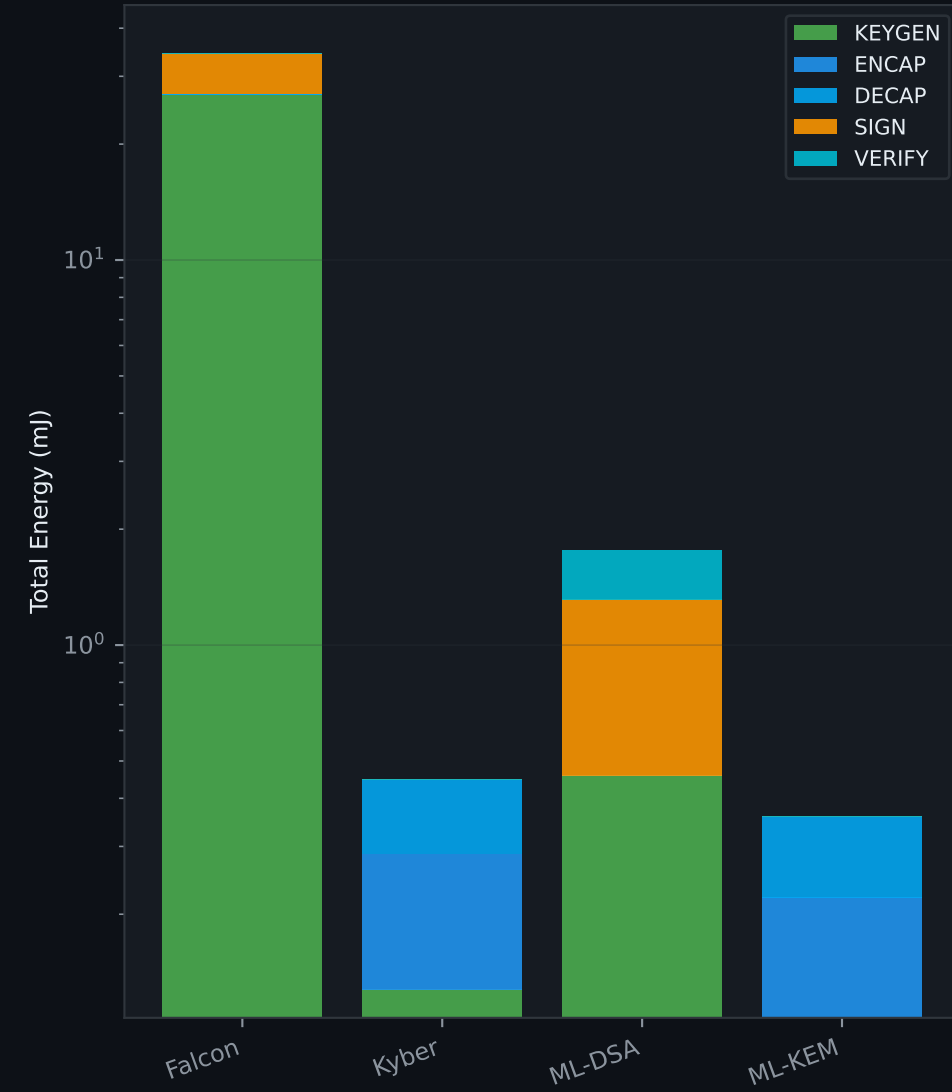


D — Energy Distribution by Algorithm Family

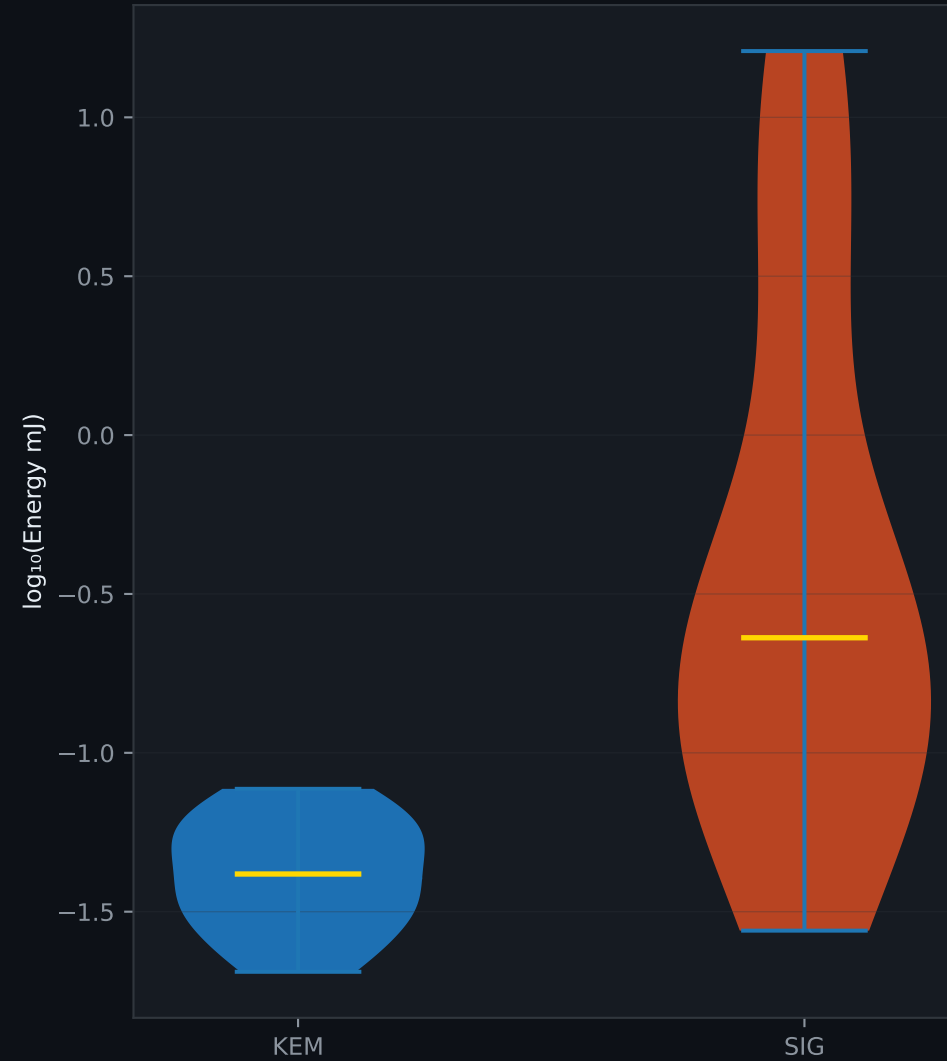


Energy Breakdown — KEM vs SIG Families

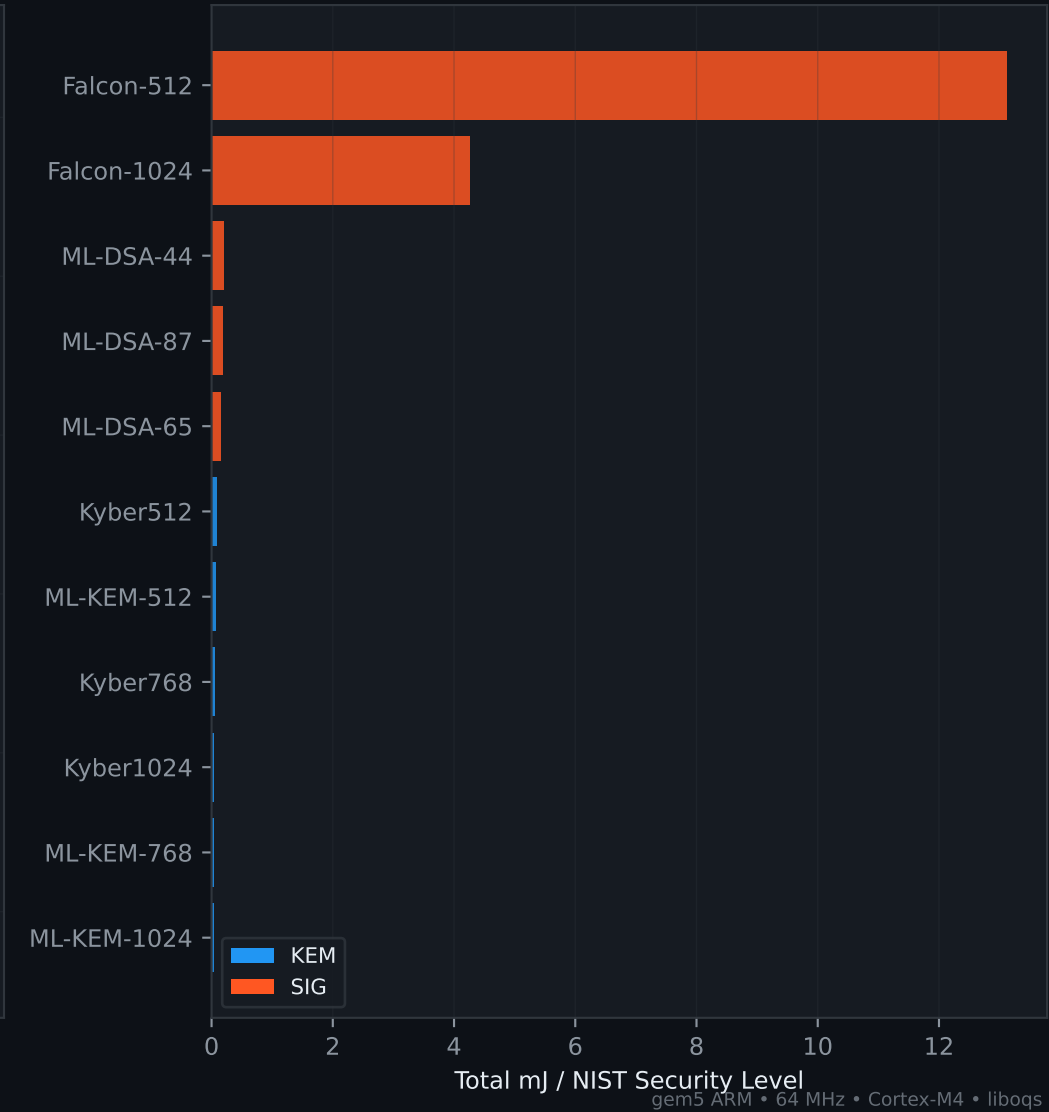
A — Total Energy by Family & Op



B — KEM vs SIG Distribution

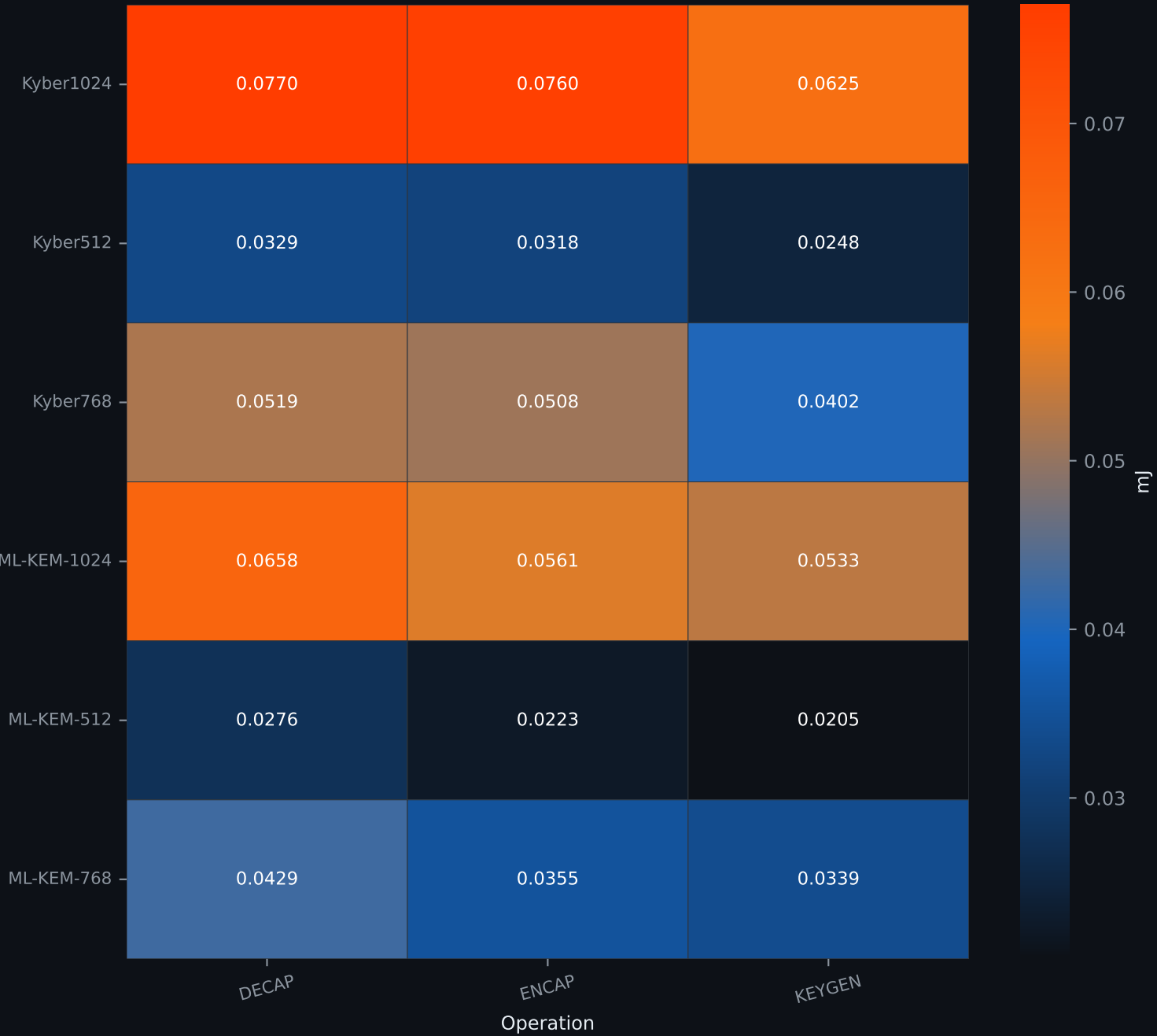


C — Energy Efficiency (lower = better per security bit)



Energy Heatmaps (mJ) — gem5 ARM Measurements

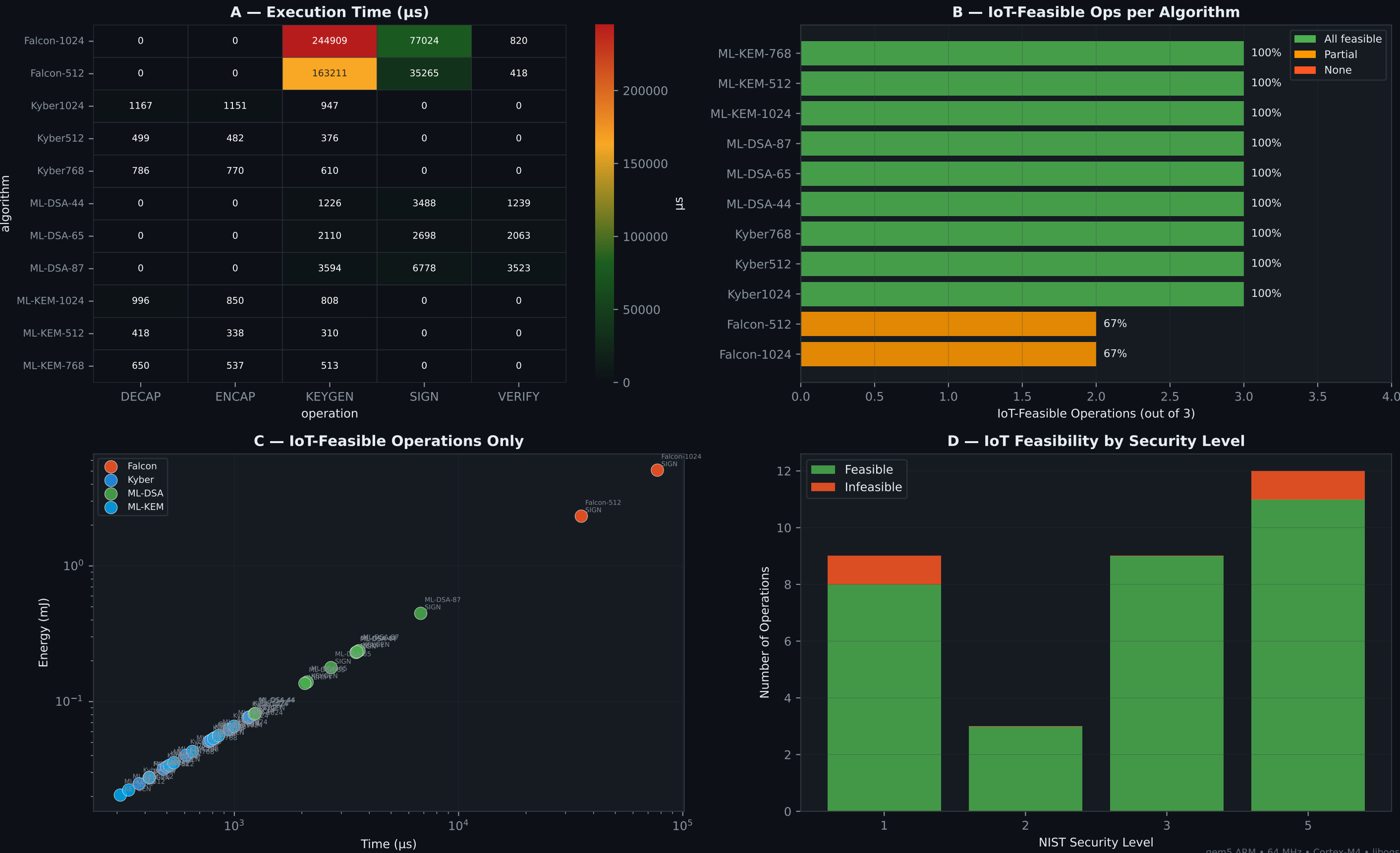
KEM Algorithms



SIG Algorithms

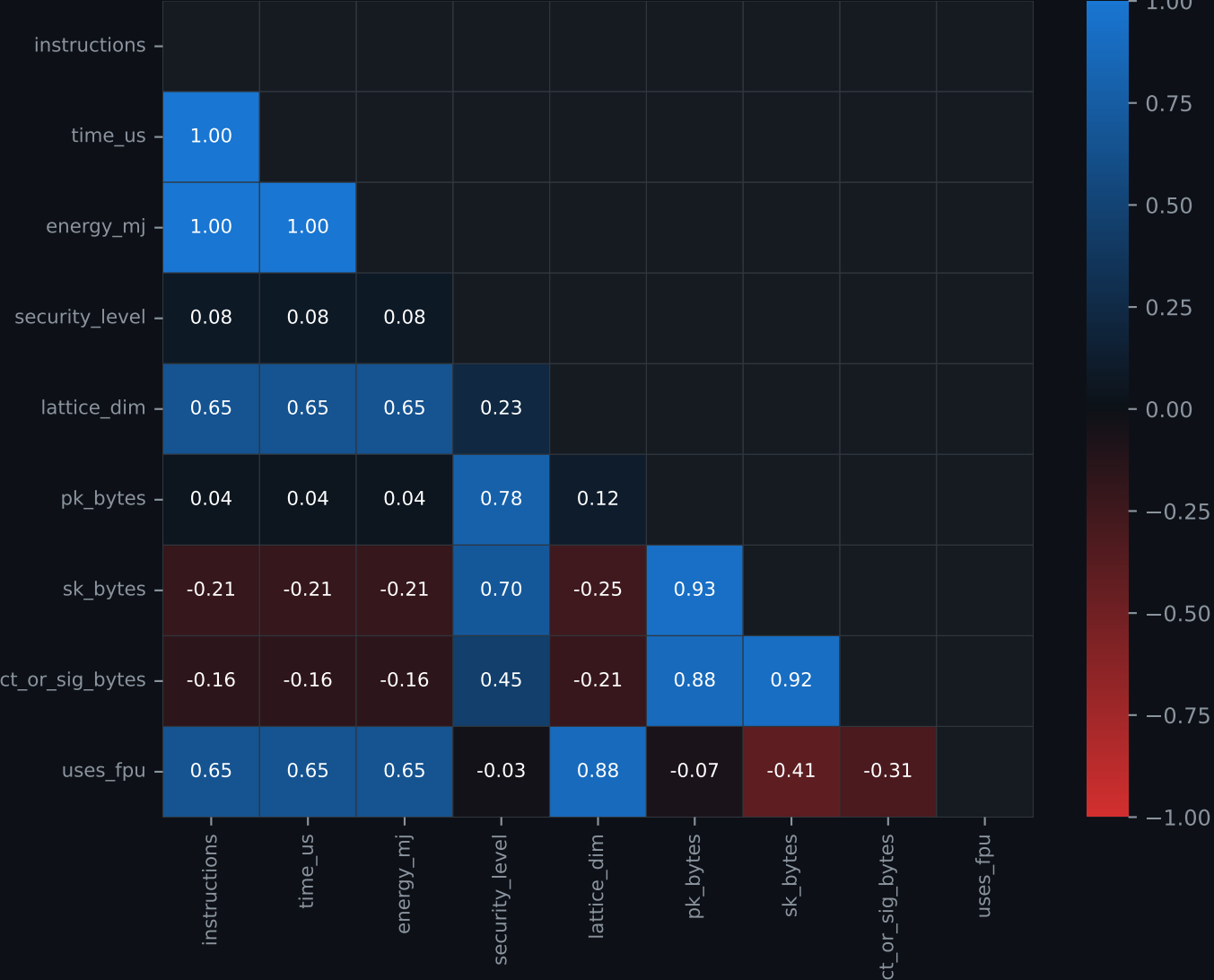


IoT Feasibility Analysis — ARM Cortex-M4 @ 64 MHz

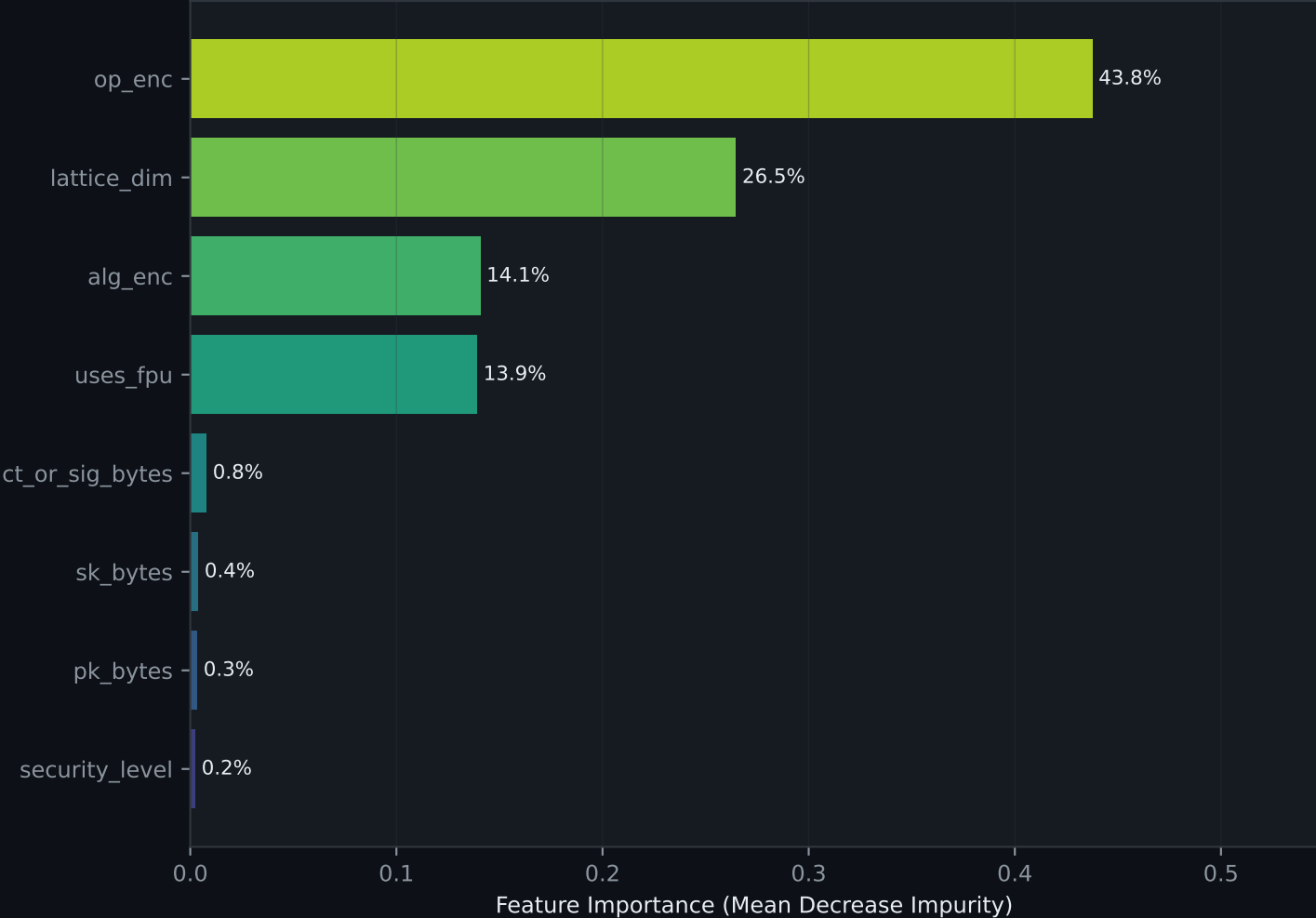


Feature Correlation & Importance for Energy Prediction

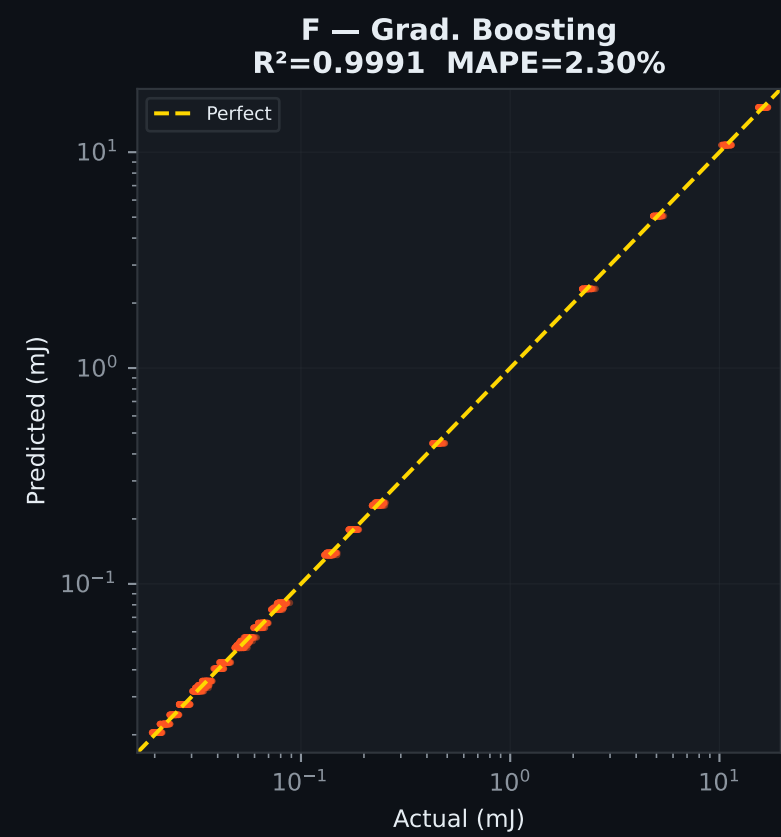
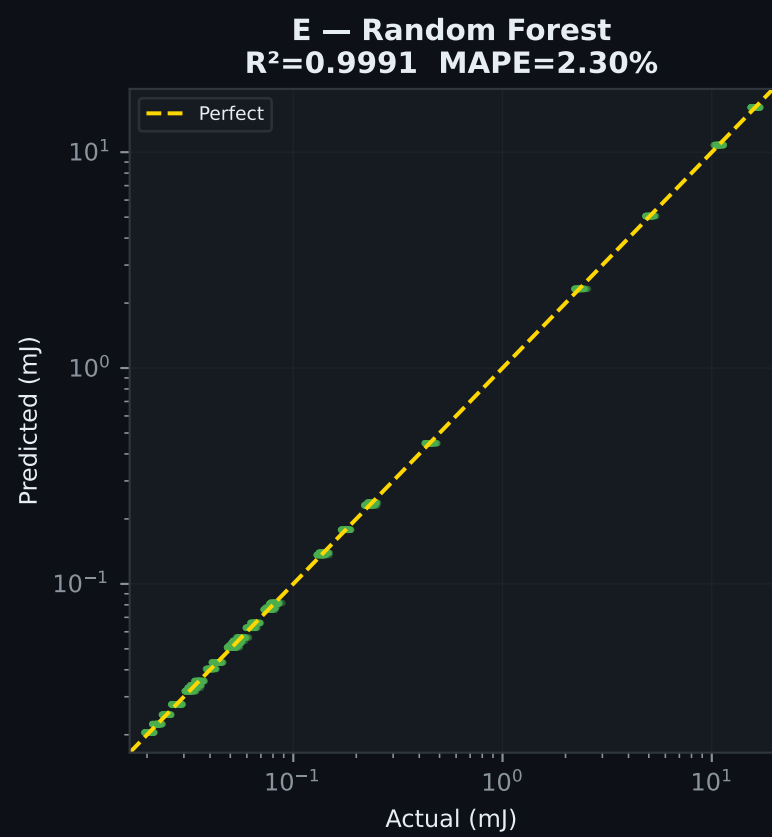
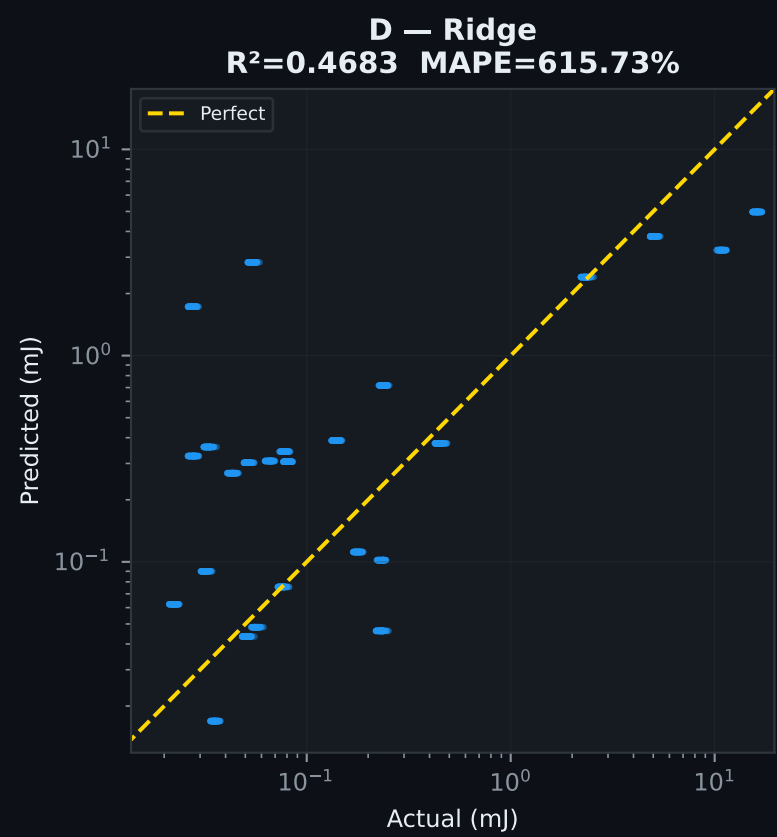
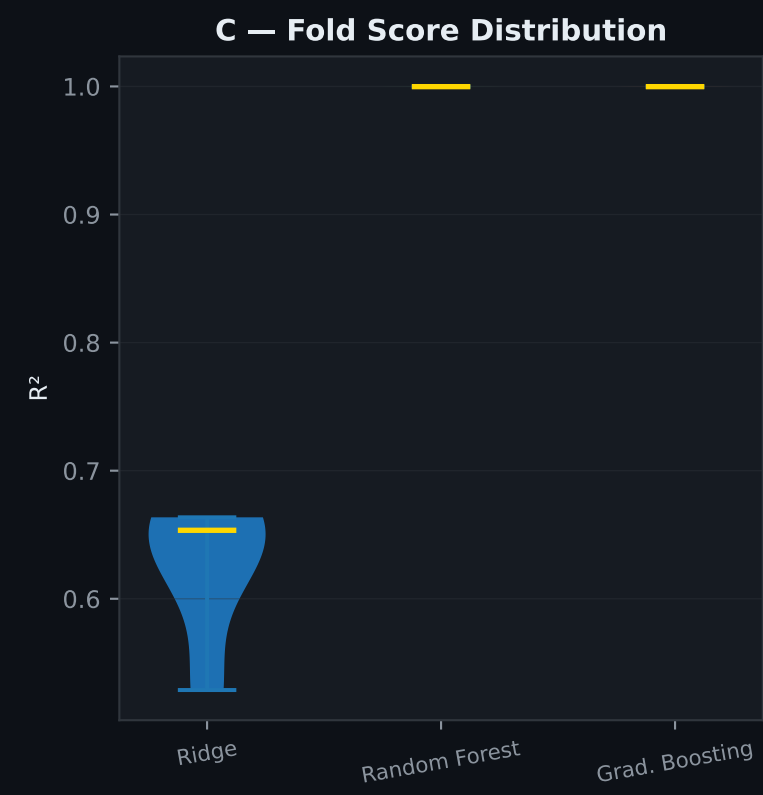
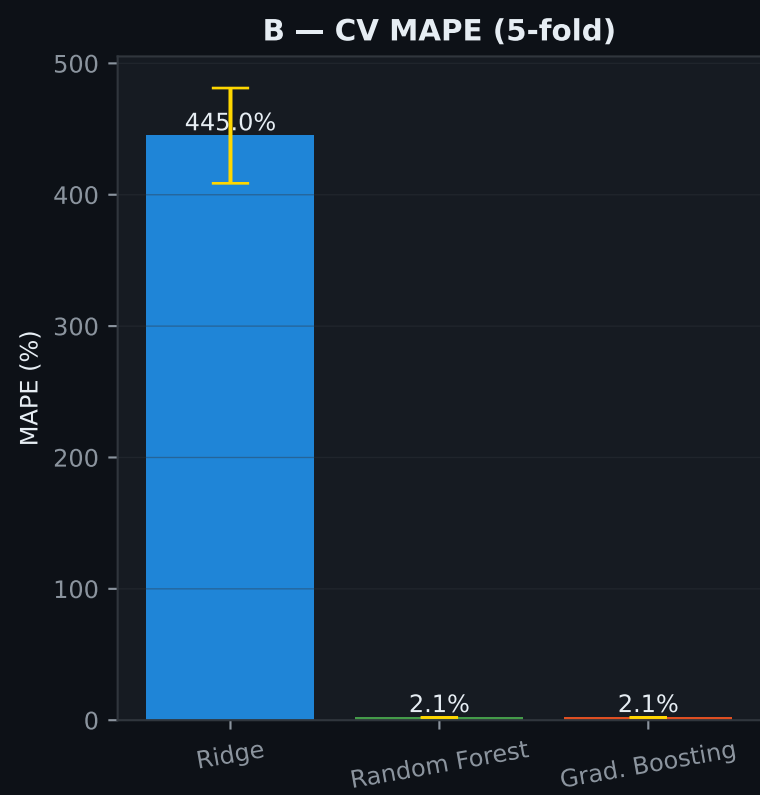
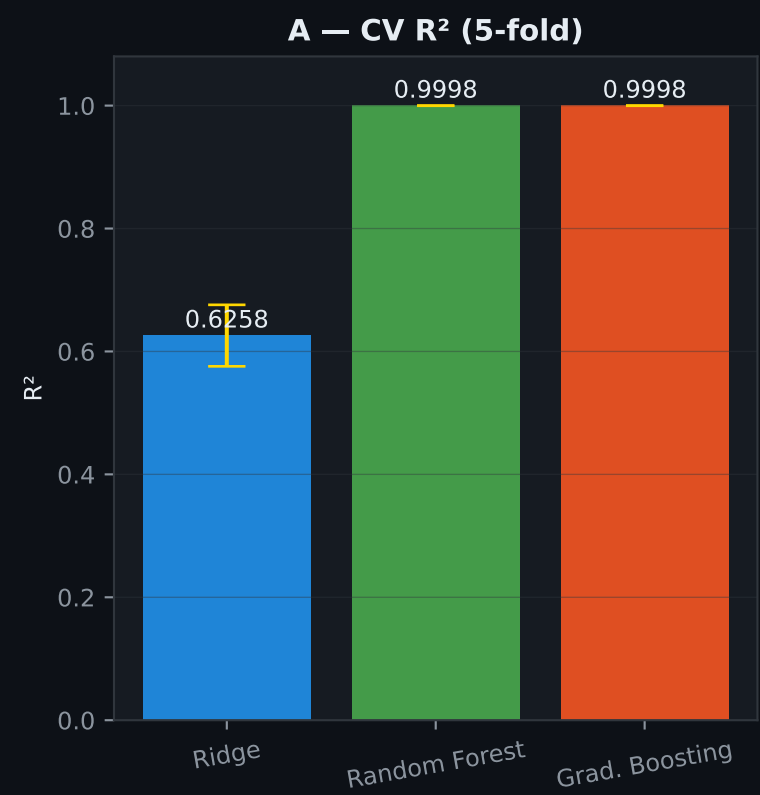
A — Pearson Correlation Matrix



**B — Random Forest Feature Importance
(predicting log energy)**

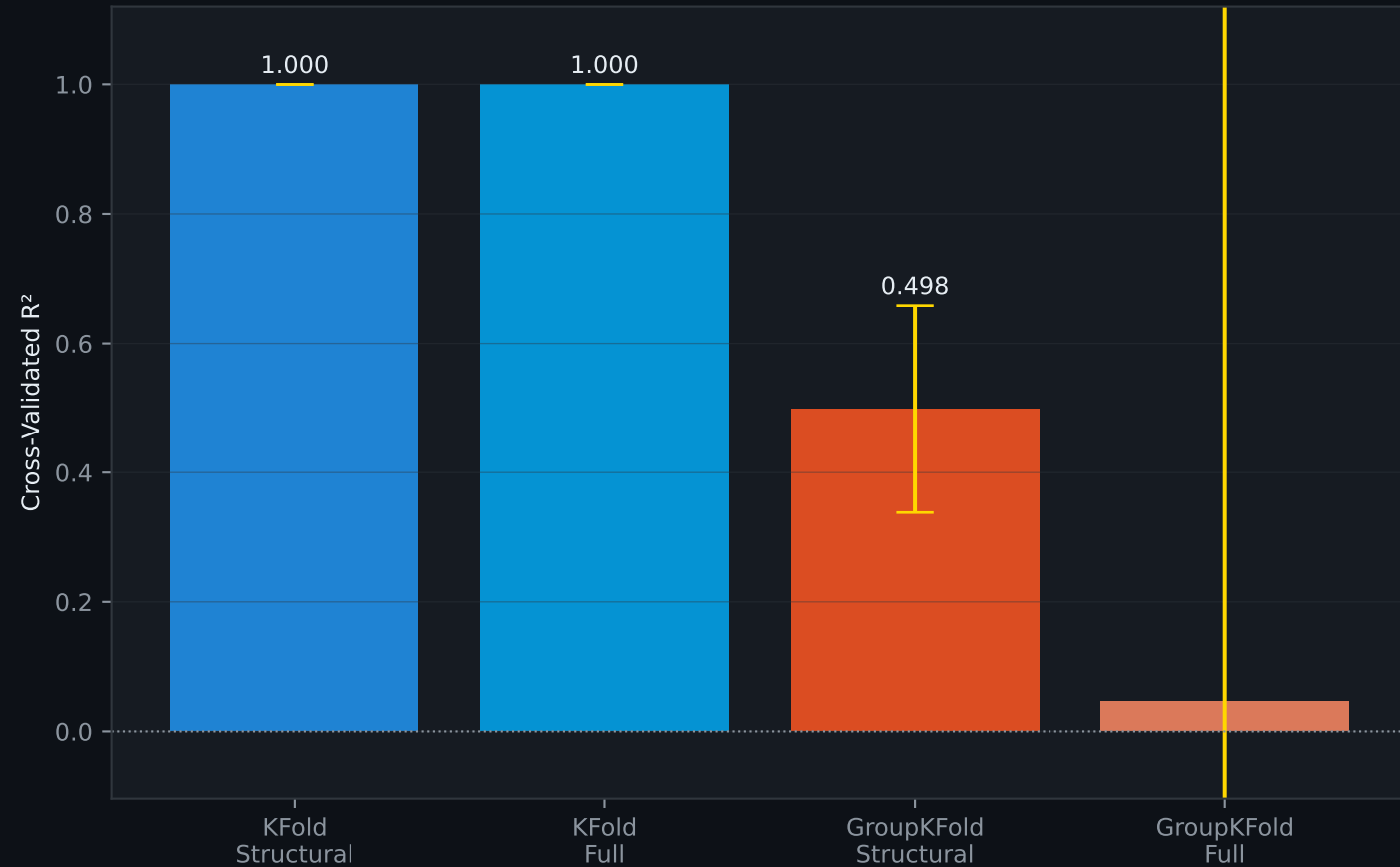


ML Energy Prediction — 5-Fold Cross Validation
Training on 100-run expanded dataset (3,300 rows)



Experiment B — Cross-Algorithm Generalisation (GroupKFold)
Tests whether model predicts unseen algorithms from structure alone

**A — KFold vs GroupKFold
(Structural vs Full features)**



**B — R^2 Fold Distribution
(width = variance)**



Summary — gem5 ARM Results & ML Model Performance

A — All gem5 Measurements (33 rows)

Algorithm	Op	Instructions	Time (µs)	Energy (mJ)	IoT
Kyber1024	DECAP	1,920,214	1167.2	0.0770	✓
Kyber1024	ENCAP	1,929,183	1151.4	0.0760	✓
Kyber1024	KEYGEN	1,574,671	947.0	0.0625	✓
Kyber512	DECAP	813,465	498.6	0.0329	✓
Kyber512	ENCAP	806,210	482.1	0.0318	✓
Kyber512	KEYGEN	619,447	376.2	0.0248	✓
Kyber768	DECAP	1,285,883	785.9	0.0519	✓
Kyber768	ENCAP	1,288,062	769.7	0.0508	✓
Kyber768	KEYGEN	1,004,095	609.7	0.0402	✓
ML-KEM-1024	DECAP	1,682,027	996.4	0.0658	✓
ML-KEM-1024	ENCAP	1,436,597	850.2	0.0561	✓
ML-KEM-1024	KEYGEN	1,379,586	807.9	0.0533	✓
ML-KEM-512	DECAP	700,646	417.5	0.0276	✓
ML-KEM-512	ENCAP	568,299	337.8	0.0223	✓
ML-KEM-512	KEYGEN	527,022	310.0	0.0205	✓
ML-KEM-768	DECAP	1,090,813	650.1	0.0429	✓
ML-KEM-768	ENCAP	903,136	537.4	0.0355	✓
ML-KEM-768	KEYGEN	871,247	513.1	0.0339	✓
Falcon-1024	KEYGEN	366,482,896	244909.3	16.1640	✗
Falcon-1024	SIGN	113,116,132	77024.4	5.0836	✓
Falcon-1024	VERIFY	1,242,518	820.4	0.0541	✓
Falcon-512	KEYGEN	251,746,781	163210.6	10.7719	✗
Falcon-512	SIGN	51,787,017	35264.5	2.3275	✓
Falcon-512	VERIFY	639,164	417.7	0.0276	✓
ML-DSA-44	KEYGEN	2,076,286	1226.2	0.0809	✓
ML-DSA-44	SIGN	5,690,037	3488.4	0.2302	✓
ML-DSA-44	VERIFY	2,064,702	1239.3	0.0818	✓
ML-DSA-65	KEYGEN	3,597,533	2110.3	0.1393	✓
ML-DSA-65	SIGN	4,472,951	2698.2	0.1781	✓
ML-DSA-65	VERIFY	3,474,117	2063.5	0.1362	✓
ML-DSA-87	KEYGEN	6,136,921	3594.0	0.2372	✓
ML-DSA-87	SIGN	11,216,615	6777.7	0.4473	✓
ML-DSA-87	VERIFY	5,968,089	3523.5	0.2326	✓

B — ML Results & IoT Feasibility

Model	CV R² Mean	± Std	CV MAPE	Note
Ridge	0.6258	± 0.0500	444.96%	
Random Forest	0.9998	± 0.0000	2.09%	★ Best
Grad. Boosting	0.9998	± 0.0000	2.10%	

IoT Feasibility Summary

✓ Kyber512

✓ Kyber768

✓ Kyber1024

✓ ML-KEM-512

✓ ML-KEM-768

✓ ML-KEM-1024

✓ ML-DSA-44

✓ ML-DSA-65

✓ ML-DSA-87

~ Falcon-512

~ Falcon-1024

3/3 ops feasible

3/3 ops feasible

3/3 ops feasible

3/3 ops feasible

3/3 ops feasible

3/3 ops feasible

3/3 ops feasible

3/3 ops feasible

3/3 ops feasible

2/3 ops feasible

2/3 ops feasible